CO1 – ASSIGNMENT 1

TASK 1 – WRITTEN ANSWER

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TASK 1 – UNDERSTAND THE PROBLEM AND IDENTIFY THE OBJECTS

Marks: 4

1.1 Objects in the Metro Smart-Card System

Object-oriented programming divides a program into objects. Each object contains data and functions that perform specific operations.

Object	Data (Attributes)	Behaviour (Functions)
Passenger	Passenger ID, passenger type	Select concession category
SmartCard	Card number, balance	Check balance, recharge and deduct fare
Journey	Starting zone, destination zone, travel time	Calculate journey fare
Zone	Zone name, base fare	Identify the fare based on the zone
FareTable	Base fares, surcharge percentage	Calculate the correct fare

Passenger stores passenger details and identifies whether the passenger is normal, a student or a senior citizen. SmartCard stores the available balance and checks whether enough money remains before deducting the fare. Journey stores the journey details needed to calculate the fare. Zone represents Inner, Middle and Outer zones. FareTable stores fare values and surcharge rules in one place.

1.2 Procedural Programming vs Object-Oriented Programming

Procedural programming: The program is divided into separate functions. Variables such as balance, fare and zone are declared separately. Functions such as calculateFare(), checkBalance() and deductFare() perform operations.

Object-oriented programming: Data and functions are grouped inside classes. For example, the SmartCard class stores the balance and provides functions to check and deduct the fare.

Procedural Programming	Object-Oriented Programming
Uses separate functions	Uses classes and objects
Data is stored separately	Data and functions are grouped
Data protection is limited	Supports data protection through encapsulation
Changes may affect many functions	Changes can be made within the related class

Object-oriented programming is more suitable for the metro system because it makes the program easier to maintain and modify.

1.3 Basic Principles of OOP

Encapsulation: Combining data and functions inside a class. The balance in SmartCard is private and can be changed using functions such as deduct() and recharge().

Abstraction: Hiding internal details and showing only the required functions. A passenger uses the metro card without knowing the internal fare calculation.

Inheritance: One class can use the properties and functions of another class. StudentPassenger and SeniorPassenger can inherit common details from Passenger.

Polymorphism: The same function name can have different behaviours. A calculateConcession() function can calculate different concessions for students and senior citizens.

1.4 Rules for a Correct Journey

• The passenger must travel within the valid metro zones.

• The journey time must be between 0 and 23 hours.

• The fare must be calculated according to the farthest zone travelled.

• A 20% surcharge is added during peak hours: 8–10 AM and 5–8 PM, with the ending times excluded.

• At least Rs. 50 must remain after fare deduction.

• The card balance must never become negative.

• Fare values must be calculated accurately.

Money should be stored in integer paise instead of decimal rupees to avoid rounding errors. The assignment does not specify the student and senior-citizen concession percentage. Therefore, the percentage must be confirmed before implementing that rule.

1.5 Alternative Concept and Complexity Analysis

A procedural solution can use separate variables and functions. The balance is stored in a variable, while calculateFare(), canTravel() and deductFare() perform the required operations.

Time complexity: Both procedural and object-oriented approaches require a fixed number of operations for each journey. Therefore, time complexity per journey is O(1).

Space complexity: Both approaches store a fixed number of values for each card. Therefore, space complexity per card is O(1).

The execution time and memory usage are almost the same. However, object-oriented programming is easier to maintain because the data and related functions are grouped inside classes.

Conclusion: Object-oriented programming is more suitable for the metro smart-card system because it provides data protection, code reusability and easier maintenance.